

Hypothesis Testing Assignment

Question 1: What is a null hypothesis (H_0) and why is it important in hypothesis testing?

Answer - The null hypothesis (H_0) is a statement that assumes there is no effect, difference, or relationship between variables. It is the starting assumption in Hypothesis Testing. It serves as the default assumption in hypothesis testing. Researchers analyze sample data to determine whether there is enough statistical evidence to reject the null hypothesis and support the alternative hypothesis.

Question 2: What does the significance level (α) represent in hypothesis testing?

Answer - The significance level (α) is the probability of rejecting the null hypothesis when it is actually true. It represents the risk of making a Type I error in hypothesis testing. It is used as a threshold to determine whether the results are statistically significant, with common values being 0.05 or 0.01.

Question 3: Differentiate between Type I and Type II errors

Answer - A Type I error occurs when a true null hypothesis is rejected, leading to a false positive result. A Type II error occurs when a false null hypothesis is not rejected, resulting in a false negative. The probability of a Type I error is represented by the significance level (α), while the probability of a Type II error is represented by β

Error Type	Meaning	Example
Type I Error	Rejecting a true null hypothesis	Concluding a medicine works when it actually doesn't
Type II Error	Failing to reject a false null hypothesis	Concluding a medicine does not work when it actually does

Question 4: Explain the difference between a one-tailed and two-tailed test. Give an example of each.

Answer- A one-tailed test examines whether a parameter is either greater than or less than a specific value, with the rejection region in one tail of the distribution.

A two-tailed test checks for any difference from a specified value, with rejection regions in both tails.

For example, testing whether a battery lasts more than 10 hours is a one-tailed test, while testing whether the average weight of a product differs from 100 grams is a two-tailed test.

Question 5: A company claims that the average time to resolve a customer complaint is 10 minutes. A random sample of 9 complaints gives an average time of 12 minutes and a standard deviation of 3 minutes. At $\alpha = 0.05$, test the claim

Answer- We will apply t-test here as population mean is not given and sample size is less than 30

Test used: **t-test** = $\bar{x} - \mu / S / \sqrt{n}$

$$t = \frac{12 - 10}{3/\sqrt{9}} = 2$$

df = 8

Critical value (p value) = ± 2.306

Since

$$|2| < 2.306$$

we fail to reject the null hypothesis (H0).

Question 6: When should you use a Z-test instead of a t-test?

Answer- A Z-test is used in Hypothesis Testing when the population standard deviation (σ) is known and the sample size is large ($n \geq 30$). It is based on the Normal Distribution

$$Z = (\bar{x} - \mu) / (\sigma / \sqrt{n}) \quad [\text{Formula for Z test}]$$

Question 7: The productivity of 6 employees was measured before and after a training program

Employee	Before	After
1	50	55
2	60	65
3	58	59
4	55	58
5	62	63
6	56	59

At $\alpha = 0.05$, test if the training improved productivity.

Answer- Let's Define the Hypothesis First:

Null hypothesis (H0) = $\mu_d = 0$

Alternative hypothesis (H1) = $\mu_d > 0$

This is a one-tailed test.

Now, Let's Calculate the difference of the productivity $d = \text{After} - \text{Before}$

Employee	Before	After	Difference
1	50	55	5
2	60	65	5
3	58	59	1
4	55	58	3
5	62	63	1
6	56	59	3

Let's get Mean of the difference

$$\bar{d} = \Sigma d / n$$

$$\bar{d} = 18 / 6 = 3$$

$$S.D = \sqrt{[\Sigma(d - \bar{d})^2 / (n - 1)]}$$

$$\Sigma(d - \bar{d})^2 = 16$$

$$S.D = \sqrt{(16 / 5)} \approx 1.789$$

* We are using a Paired T test for this

$$t = \bar{d} / (sd / \sqrt{n})$$

$$t = 3 / (1.789 / \sqrt{6})$$

$$t \approx 4.11$$

The P value

$$n = 6$$

Degree of Freedom

$$df = n - 1 = 5$$

From T table at $\alpha = 0.05$

$$t_{\text{critical}} \approx 2.015$$

The Decision Values

$$t_{\text{calculated}} = 4.11$$

$$t_{\text{critical}} = 2.015$$

Conclusion

At $\alpha = 0.05$, the calculated t-value is greater than the critical value. Therefore, the null hypothesis is rejected. This indicates that the training program **significantly improved employee productivity**.

Question 8: A company wants to test if product preference is independent of gender.

Gender	Product A	Product B	Total
Male	30	20	50
Female	10	40	50
Total	40	60	100

At $\alpha = 0.05$, test independence

Answer- A Chi-Square Test of Independence is used to test whether product preference is independent of gender. (as question provides frequency not Numeric so we use chi square test)

Hypotheses

H_0 : Product preference is independent of gender

H_1 : Product preference is dependent on gender

Expected frequencies

Expected value formula (E) = (Row Total $\times$ Column Total) / Grand Total

$$\text{Male-Product A} = (50 \times 40) / 100 = 20$$

$$\text{Male-Product B} = (50 \times 60) / 100 = 30$$

$$\text{Female-Product A} = (50 \times 40) / 100 = 20$$

$$\text{Female-Product B} = (50 \times 60) / 100 = 30$$

Chi-square statistic

$$\chi^2 = \sum (O - E)^2 / E$$

$$\chi^2 = 5 + 3.33 + 5 + 3.33 = \mathbf{16.66}$$

Degrees of freedom

$$df = (r-1)(c-1) = (2-1)(2-1) = 1$$

At $\alpha = 0.05$, $\chi^2_{\text{critical}} = \mathbf{3.841}$

Since $\mathbf{16.66} > \mathbf{3.841}$, the null hypothesis is rejected

Conclusion:

Product preference is **not independent of gender**, indicating that gender influences the choice between Product A and Product B